# Arista Access Node KR-ACC-24 Hardware/Environmental Fault Failure Resolution SOP

**Node:** KR-ACC-24 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_6219

## Description

This Standard Operating Procedure is designed to address hardware/environmental fault failures on Arista Access Node KR-ACC-24. Specifically, this SOP targets ASIC memory parity errors, line card crashes, and power supply redundancy loss, which are critical failure domains within the Access Node. The described protocol and hardware mechanics governing these faults are inherent to the Arista EOS (Extensible Operating System) and the underlying hardware architecture. The ASPM (Active Serialization and Power Management) and the EOS's line card redundancy protocol govern the data and control plane operations. The failure of the power supply redundancy, in particular, triggers an alarm code (ALARM_6219) with a warning severity, necessitating immediate attention to prevent potential packet loss, network instability, or even more severe hardware failures. This SOP will guide the resolution of the fault, ensuring minimal downtime and maximum network availability.

## Pre-requisite Checks

- Check: Verify the site location and Node ID
- Check: Confirm the power supply redundancy status
- Check: Ensure all remote hands are dispatched to KR Puram Access
- Check: Verify the redundant link status

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity errors | High |
| Line card crashes | Medium |
| Power supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy status: active, standby: ready, Description: This command verifies the power supply redundancy status, ensuring the active and standby units are ready and functioning correctly.

Command: show hardware, Expected Output: ASIC memory parity errors: 0, Description: This command displays the hardware status, specifically checking for ASIC memory parity errors, which could indicate a
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hardware fault.

Command: show system log, Expected Output: Line card crashes: recent crash logs, Description: This command retrieves the system log, searching for recent line card crashes or other error messages that may indicate a hardware or software issue.
```

## Resolution Steps

Step 1: Step: Power cycle the Node, Command: system power-cycle, Description: This command initiates a power cycle of the Node, which will reboot the system and potentially resolve the fault.

Step 2: Step: Swap out the faulty power supply, Command: system redundancy swap-supply, Description: This command enables the standby power supply to become active, replacing the faulty unit and resolving the redundancy loss.

Step 3: Step: Replace the faulty line card, Command: system replace line-card, Description: This command initiates the replacement of the faulty line card with a new one, ensuring the Node's functionality is restored.

## Post-Check Verification

```
Command: show system status, Expected Output: Node status: up, power supply: active, line card: up, Description: This command verifies the Node's status, ensuring it is up and functioning correctly with the active power supply and operational line card.

Command: show system log, Expected Output: No recent crashes or errors, Description: This command checks the system log for any recent crashes or errors, ensuring the fault has been resolved and the Node is stable.
```

## Roll-back Procedures

- Step: Restore the original power supply, Command: system redundancy restore-supply, Description: This command reverts the power supply to its original state, replacing the standby unit with the faulty one. This is done to roll back the changes made during the resolution process.

- Step: Revert the line card replacement, Command: system revert line-card, Description: This command reverts the line card replacement, restoring the faulty unit. This is done to roll back the changes made during the resolution process.